

How to Trace an Email



SIMPLE

mAIL

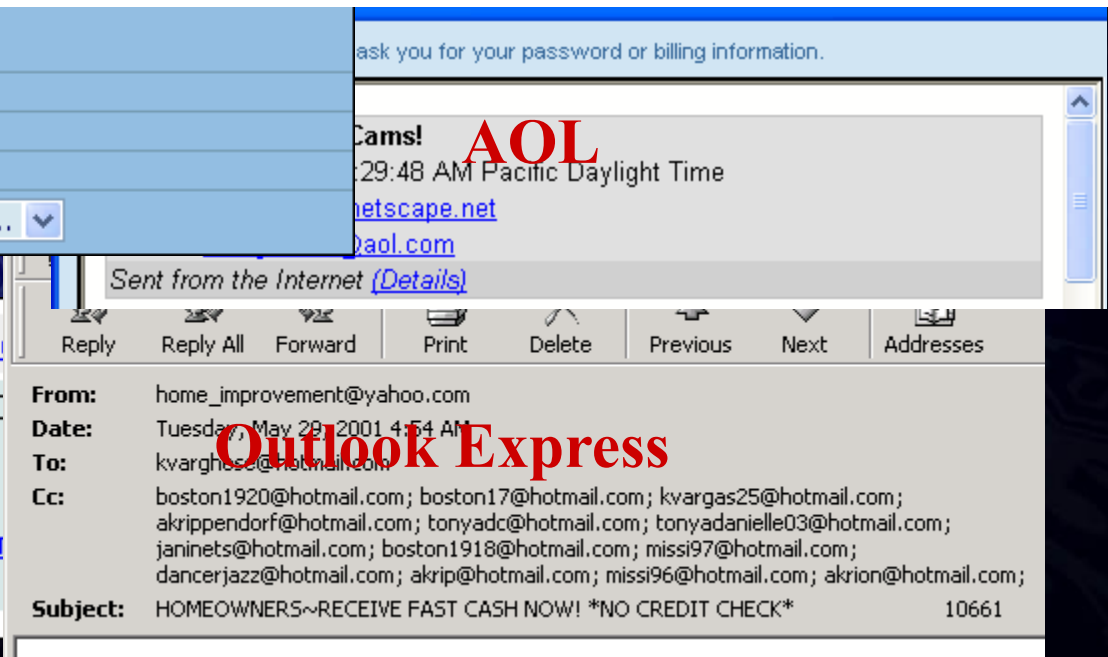
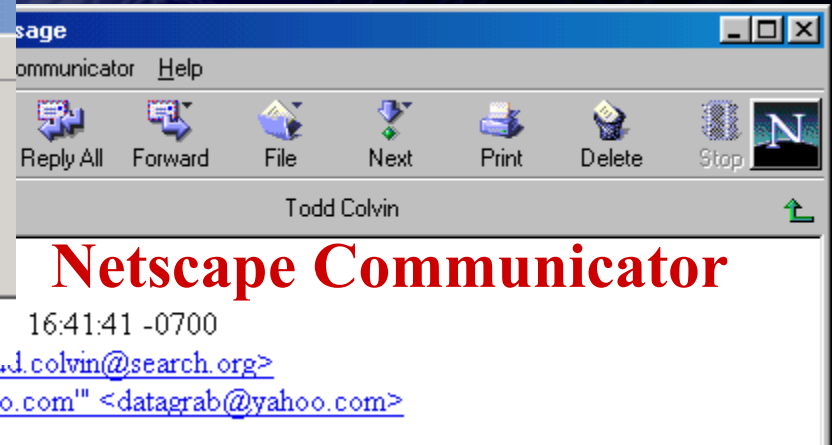
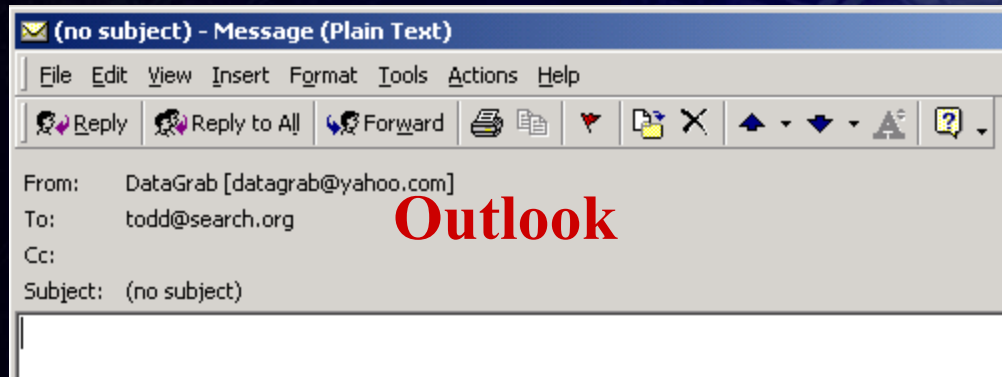
tRANSFER

pROTOCOL

Email Tracing

- In order to determine the sender of an email, an investigator needs the email's header information.
- An email header is the information added to the beginning/top of the electronic message.
- By default, email clients and services only show an abbreviated form of the header.

Abbreviated Headers



Email Tracing

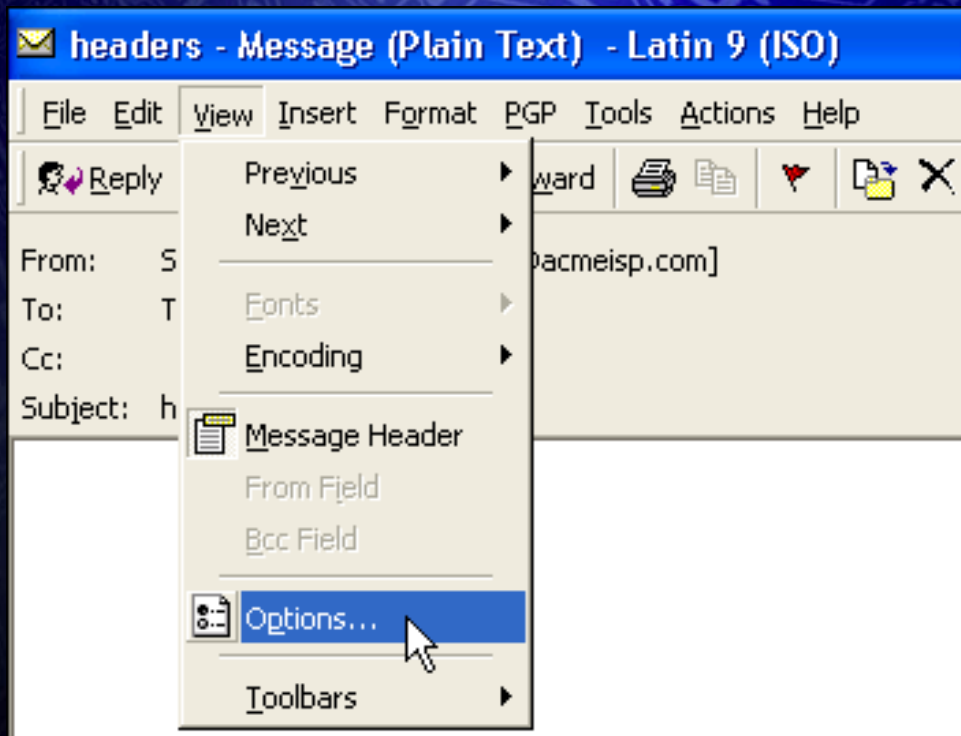
- Information displayed in the abbreviated view are the memo header items:
 - From
 - To
 - CC
 - Subject
 - Date
- This is not enough information to properly trace an email.

Email Tracing

- The complete email header is available through additional steps.
- The necessary steps vary, depending on the type of email client used. For example:
 - Outlook
 - Outlook Express
 - AOL
 - Eudora
 - Entourage
 - And others have different procedures.

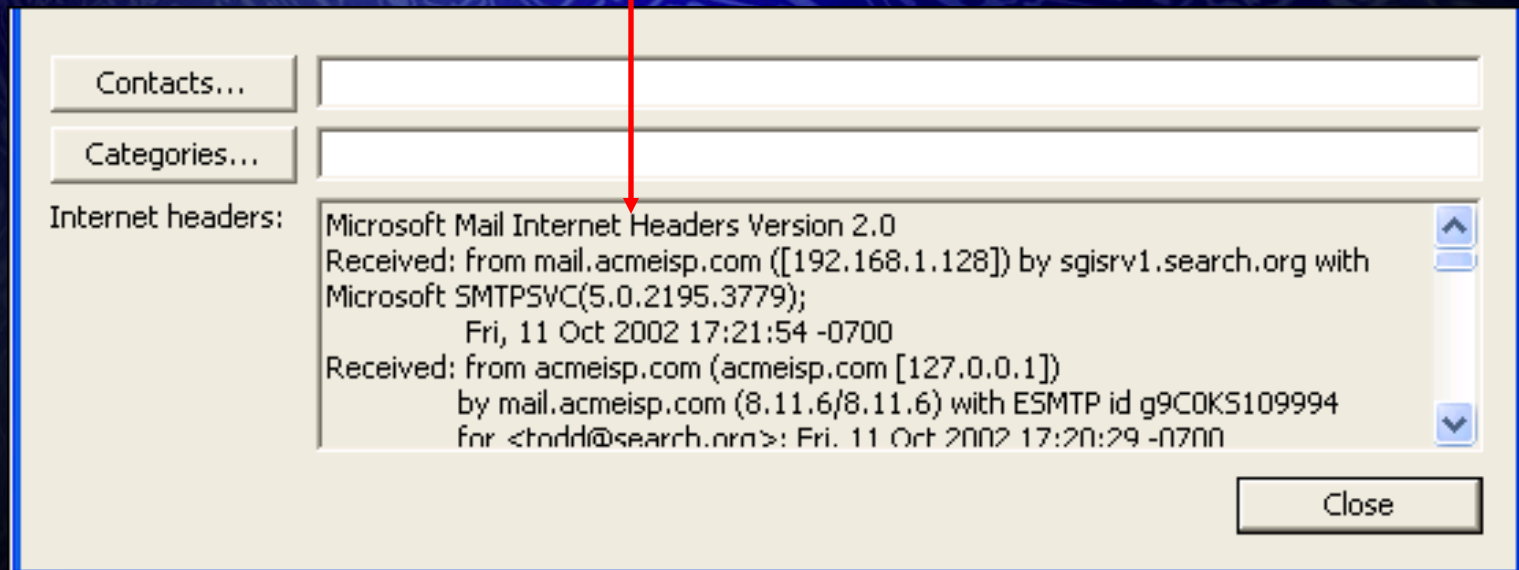
Email Tracing

- Example of viewing the complete header in Outlook 2000.
 - Open the email, click on the View menu, select Options...



Email Tracing

- Example of viewing the complete header in Outlook 2000.
 - The complete header is at the bottom of the window that appears.



Email Tracing

- Example of a complete email header:

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)

From: John Doe <johndoe@yahoo.com>

Subject: Fwd: Free kittens?

To: jqs@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]

Email Tracing

- The complete email header is a record of:
 - Who sent the email.
 - Which network it originated from.
 - Which email servers processed it.
 - Miscellaneous information:
 - Timestamps
 - Email client
 - Encoding information, etc.
 - Some of it is useful for tracing email, some of it is not.

Email Tracing

- Headers are created by the email servers that process messages for delivery.
- Not every server adds detailed information to the header.
 - Depends on the email protocol used.
- Headers of the type seen in the previous example are created by servers using the Simple Mail Transfer Protocol (SMTP) to transfer email to its destination.
 - SMTP is the protocol for transmitting email across the Internet.

Email Tracing

- Other common protocols used for email:
 - Post Office Protocol (POP)
 - Internet Message Access Protocol (IMAP)
 - MAPI (Messaging Application Programming Interface)
- These protocols are used for communication between the user's email program and the user's email server.
 - They are not the protocols used for transferring (relaying) emails across the Internet between email servers.

Email Tracing

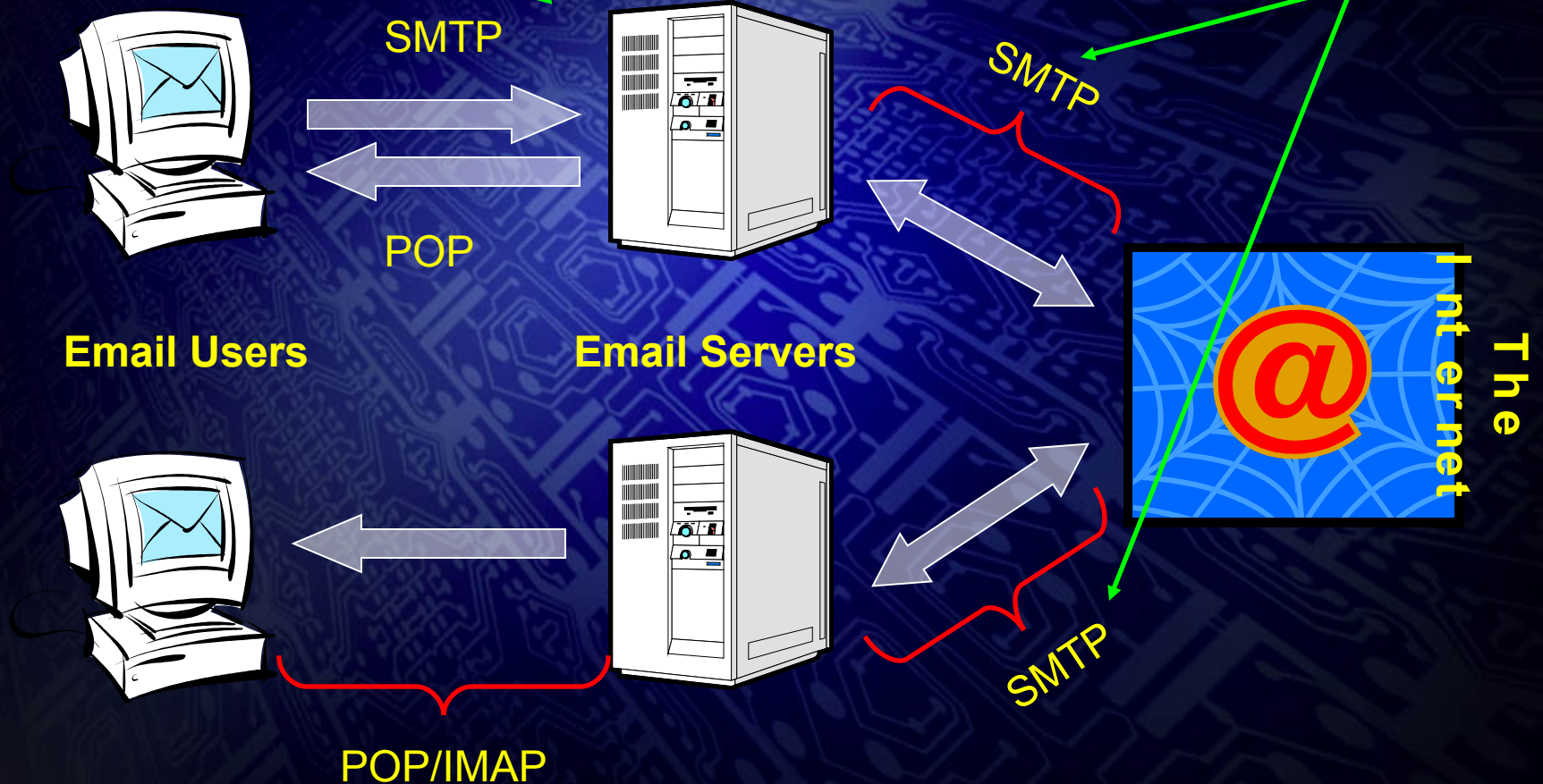
- Can combine the use of two protocols.
 - Common among ISPs for home users or when outsourcing email services.

In
Out



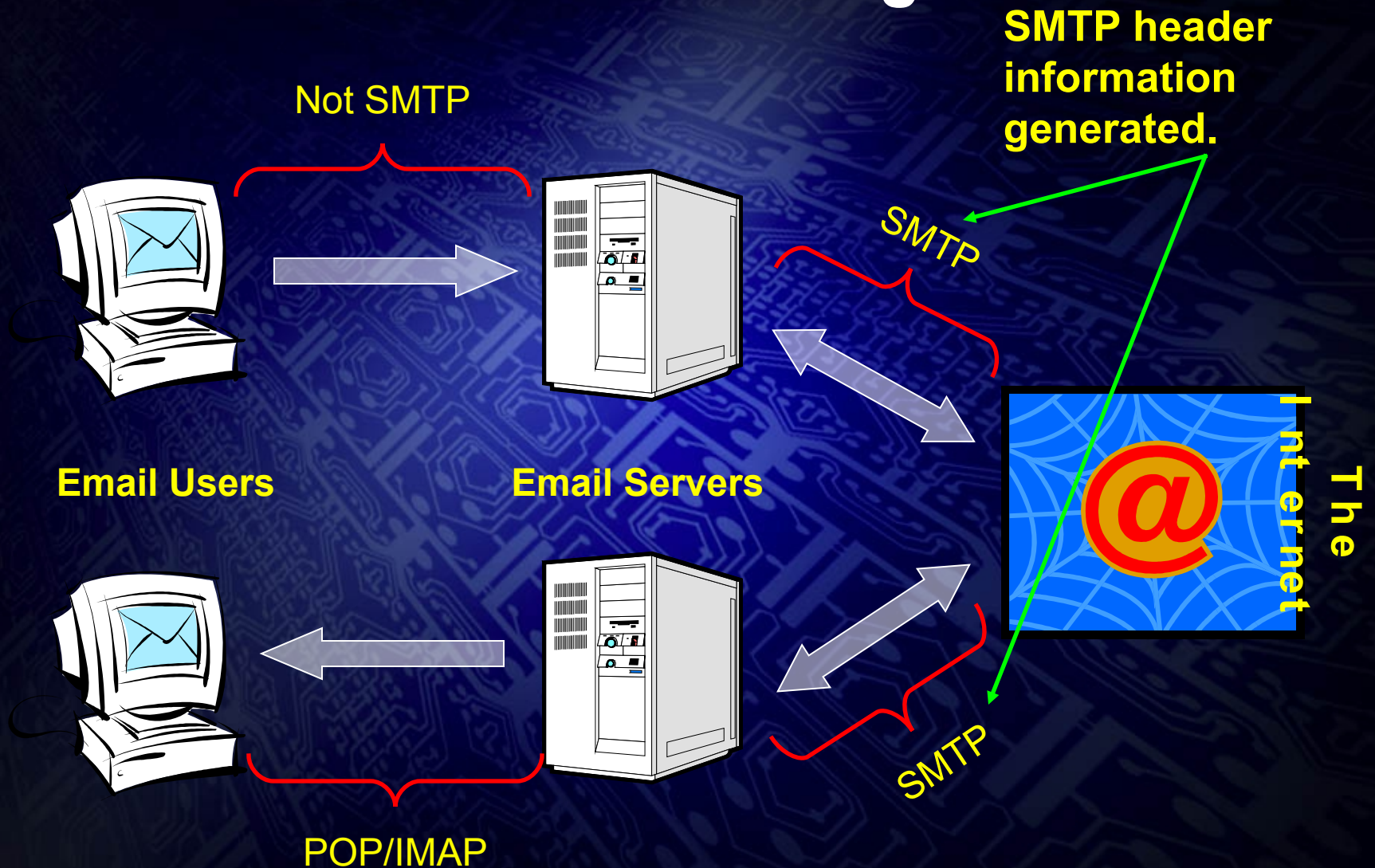
Email Tracing

**SMTP header
information
generated.**



Example of common email transmission and use of protocols

Email Tracing

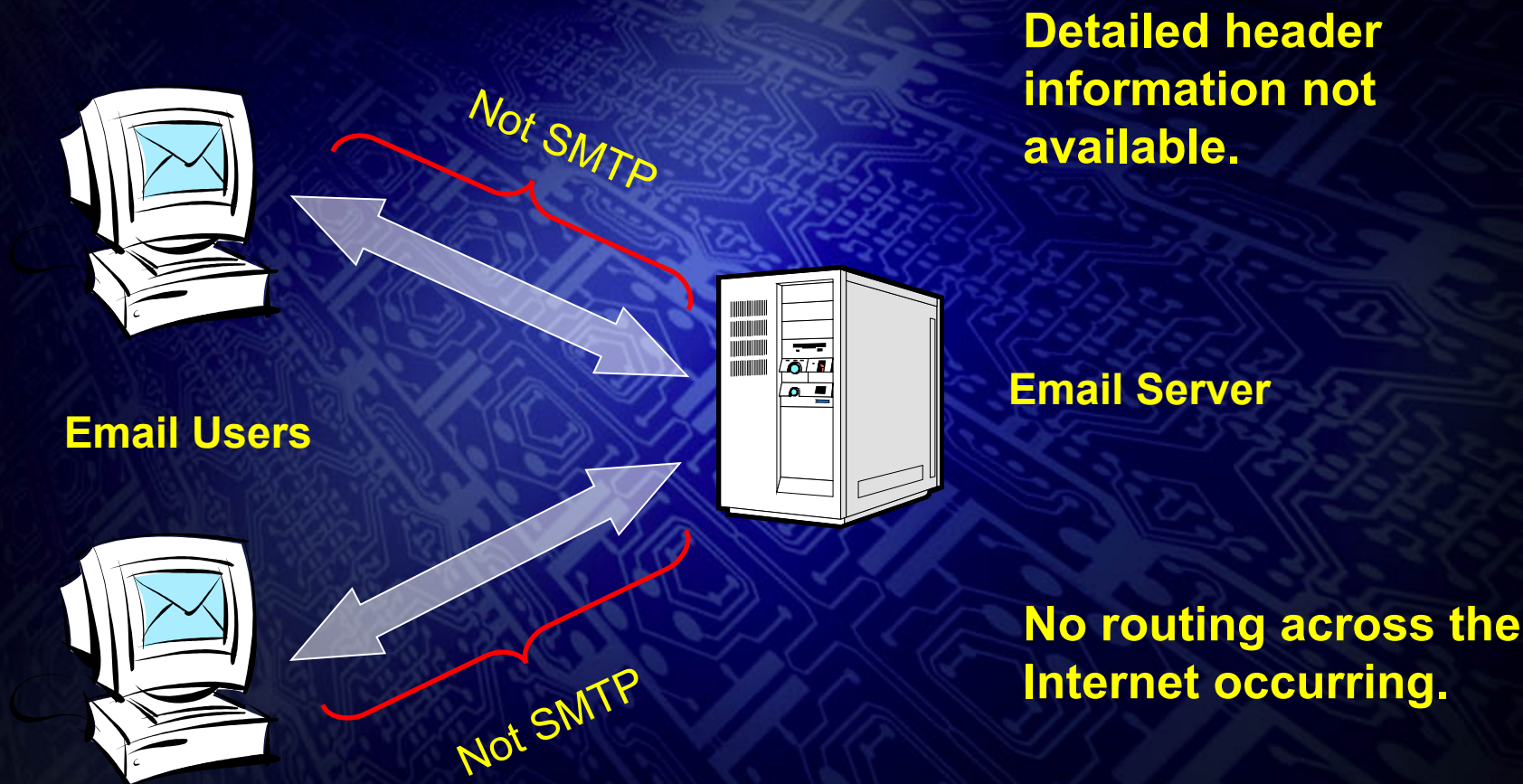


Example of common email transmission and use of protocols

Email Tracing

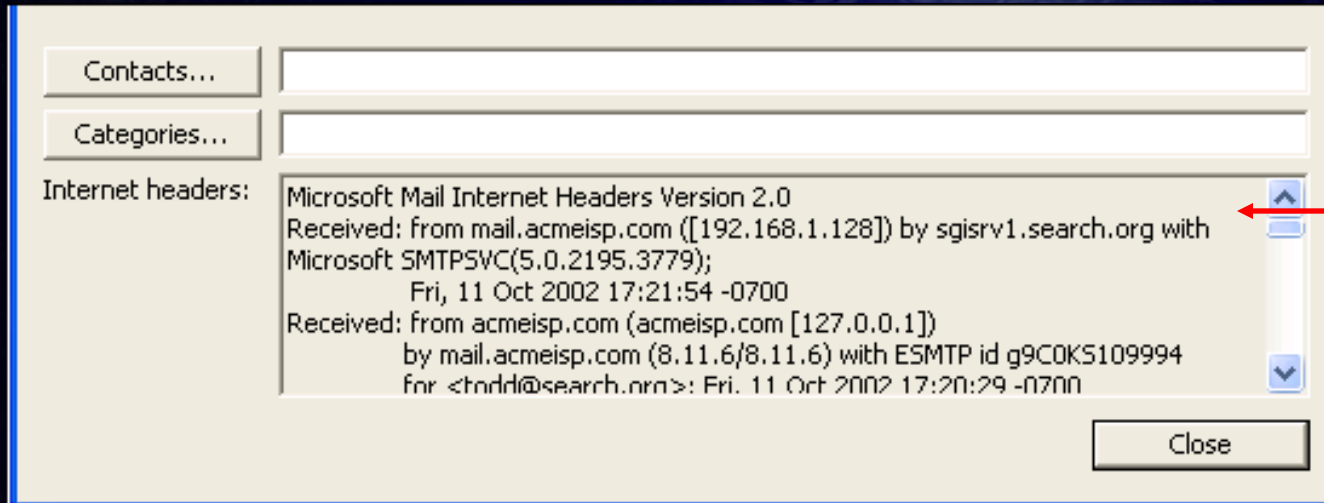
- Email systems not using the SMTP will not have the same detailed information available as those transmitted with SMTP.
 - Non-SMTP transmissions are common when email is exchanged among people using the same email system as the sender, for example:
 - AOL user to AOL user
 - Between colleagues (coworkers) at work.

Email Tracing



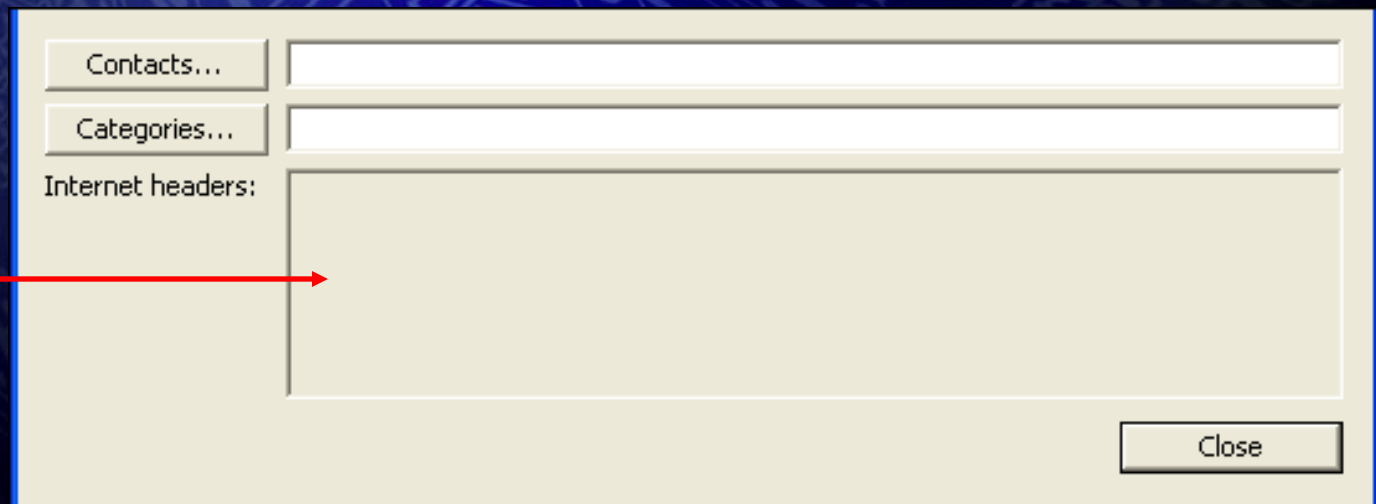
Example of email sent between users of the same email system without SMTP

Email Tracing

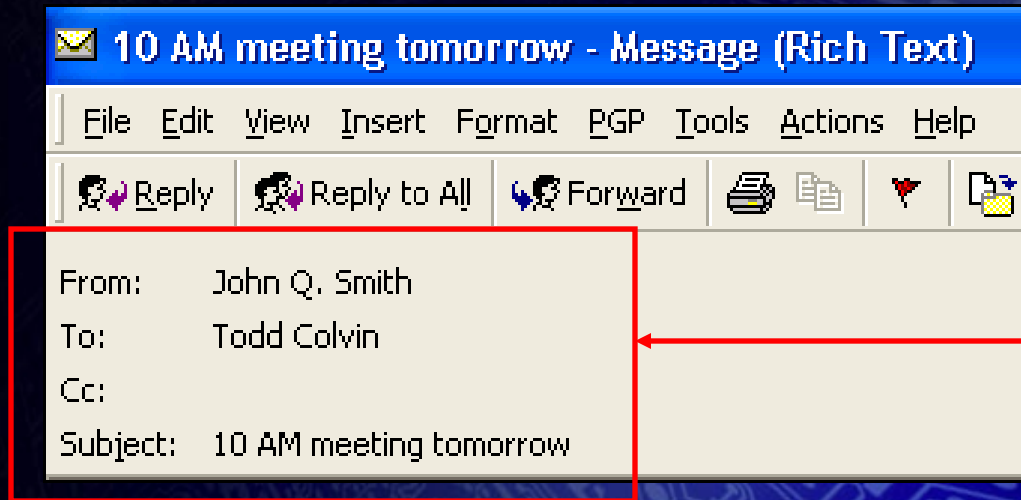


**With
SMTP**

**Without
SMTP**



Email Tracing



Memo header items only
in a non-SMTP
environment.

Email Header Details

- Headers are comprised of lines of information called header fields.
- Each field contains a field label, followed by a colon “:” and then the field body.

From: petshopboys@aol.com



Field label Field body

Email Header Details

- RFC 2076 – Common Internet Message Headers
 - A table of commonly occurring header fields and a short description of what they mean and what their status is (e.g., standard or not).
 - Not exhaustive.
 - <http://www.ietf.org/rfc/rfc2076.txt>

Email Header Details

- No limits on the number of fields.
 - Senders can create fields.
- Only two types are required:
 - Origination date field “Date:”
 - At least one of the originator fields:
 - From:
 - Sender:
 - Reply-To:

Email Header Details

- Header fields of particular interest:
 - The originator fields -
 - From:, Sender:, Reply-To:
 - Date:
 - Received:
 - X-Originating-IP:
 - Message-ID:
 - X-Mailer:
 - X-MIMEOLE:

Email Tracing

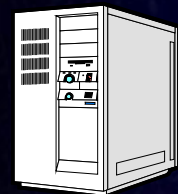
- SMTP email header structure:
 - The fields are loosely organized in a layered, bottom-to-top sequence.
 - Few limitations on the order.
 - Mostly apply to certain fields (e.g., Received and Return-Path).
 - First field is on the bottom, subsequent fields added on top, in the order they are written.



Email Tracing

- How many email servers a message will pass through depends on the networks it passes through.
- Then the third email server adds information and so on until the destination is reached. ← 3...
- The second email server adds information next. ← 2
- The first email server to receive the message via SMTP is the first to add information into the header. ← 1

Email Tracing



**First
Email Server**

The first email server receiving the message via SMTP will create detailed header information of this.



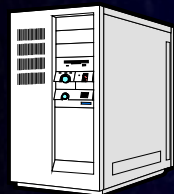
SMTP



Email User

Client is sending email to the server via SMTP.

Email Tracing



**First
Email Server**



SMTP

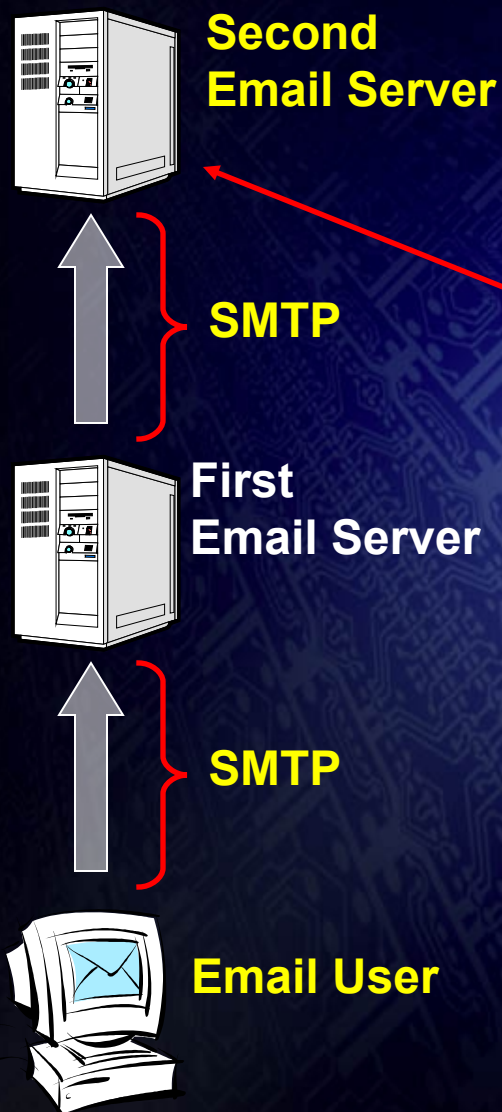


Email User

Received: from jqs.search.org ([192.168.1.143]) by sgisrv1.search.org with
Microsoft SMTPSVC(5.0.2195.5329); Wed, 23 Oct 2002 18:02:28 -0700
Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>
From: "John Q. Smith" jqs@search.org
To: datagrab@yahoo.com
Subject: Kittens for sale
Date: Wed, 23 Oct 2002 17:56:38 -0700
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="-----
=_NextPart_000_001C_01C27ABD.890FD900"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000

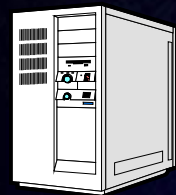
The actual fields added will vary.

Email Tracing



If the message is relayed again, the second email server to receive the message via SMTP will add another Received field.

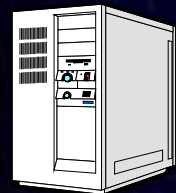
Email Tracing



**Second
Email Server**



SMTP



**First
Email Server**



SMTP



Email User

Received: from 64.162.18.191 (EHLO sgisrv1.search.org) (64.162.18.191) by mta629.mail.yahoo.com with SMTP; 23 Oct 2002 18:02:30 -0700 (PDT)

Received: from jqs.search.org ([192.168.1.143]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.5329); Wed, 23 Oct 2002 18:02:28 -0700

Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>

From: "John Q. Smith" jqs@search.org

To: datagrab@yahoo.com

Subject: Kittens for sale

Date: Wed, 23 Oct 2002 17:56:38 -0700

MIME-Version: 1.0

Content-Type: multipart/alternative; boundary="-----
_NextPart_000_001C_01C27ABD.890FD900"

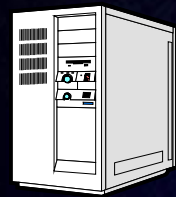
X-Priority: 3

X-MSMail-Priority: Normal

X-Mailer: Microsoft Outlook Express 6.00.2600.0000

X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000

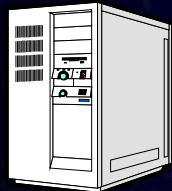
Email Tracing



**Second
Email Server**



SMTP



**First
Email Server**



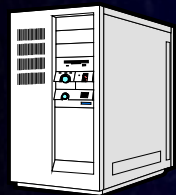
SMTP



Email User

X-Apparently-To: datagrab@yahoo.com via 216.136.173.224; 23 Oct 2002 18:02:30 -0700 (PDT)
Return-Path: <jqs@search.org>
Received: from 64.162.18.191 (EHLO sgisrv1.search.org) (64.162.18.191) by mta629.mail.yahoo.com with SMTP; 23 Oct 2002 18:02:30 -0700 (PDT)
Received: from jqs.search.org ([192.168.1.143]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.5329); Wed, 23 Oct 2002 18:02:28 -0700
Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>
From: "John Q. Smith" jqs@search.org
To: datagrab@yahoo.com
Subject: Kittens for sale
Date: Wed, 23 Oct 2002 17:56:38 -0700
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="-----_NextPart_000_001C_01C27ABD.890FD900"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000
Return-Path: jqs@search.org
X-OriginalArrivalTime: 24 Oct 2002 01:02:28.0329 (UTC)
FILETIME=[05D81D90:01C27AF9]
Content-Length: 662

Email Tracing



**First
Email Server**



**Not
SMTP**

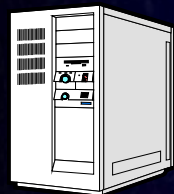


Email User

No detailed header information yet. Just basic memo header items.

Client is sending email to the server via IMAP.

Email Tracing



**First
Email Server**



SMTP

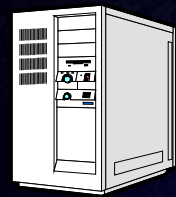


Email User

Received: from jqs.search.org ([192.168.1.143]) by sgisrv1.search.org with
Microsoft SMTPSVC(5.0.2195.5329); Wed, 23 Oct 2002 18:02:28 -0700
Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>
From: "John Q. Smith" jqs@search.org
To: datagrab@yahoo.com
Subject: Kittens for sale
Date: Wed, 23 Oct 2002 17:56:38 -0700
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="-----
=_NextPart_000_001C_01C27ABD.890FD900"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000

The actual fields added will vary.

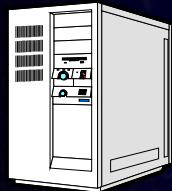
Email Tracing



**Second
Email Server**



SMTP



**First
Email Server**



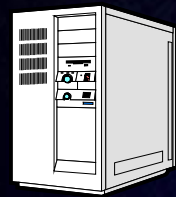
**Not
SMTP**



Email User

Received: from 64.162.18.191 (EHLO sgisrv1.search.org) (64.162.18.191) by
mta629.mail.yahoo.com with SMTP; 23 Oct 2002 18:02:30 -0700 (PDT)
Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>
From: "John Q. Smith" jqs@search.org
To: datagrab@yahoo.com
Subject: Kittens for sale
Date: Wed, 23 Oct 2002 17:56:38 -0700
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="-----
_NextPart_000_001C_01C27ABD.890FD900"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000

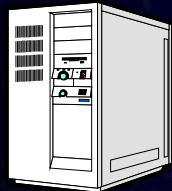
Email Tracing



**Second
Email Server**



SMTP



**First
Email Server**



IMAP



Email User

X-Apparently-To: datagrab@yahoo.com via 216.136.173.224; 23 Oct 2002 18:02:30 -0700 (PDT)
Return-Path: <jqs@search.org>
Received: from 64.162.18.191 (EHLO sgisrv1.search.org) (64.162.18.191) by mta629.mail.yahoo.com with SMTP; 23 Oct 2002 18:02:30 -0700 (PDT)
Message-ID: <002901c27af8\$521b8ad0\$8f01a8c0@sgisrv1.search.org>
From: "John Q. Smith" jqs@search.org
To: datagrab@yahoo.com
Subject: Kittens for sale
Date: Wed, 23 Oct 2002 17:56:38 -0700
MIME-Version: 1.0
Content-Type: multipart/alternative; boundary="====_NextPart_000_001C_01C27ABD.890FD900"
X-Priority: 3
X-MSMail-Priority: Normal
X-Mailer: Microsoft Outlook Express 6.00.2600.0000
X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000
Return-Path: jqs@search.org
X-OriginalArrivalTime: 24 Oct 2002 01:02:28.0329 (UTC)
FILETIME=[05D81D90:01C27AF9]
Content-Length: 662

Email Tracing

- With headers created by SMTP servers, there are 3 primary header fields used to trace emails:
 - Sender's email address ← Clearly labeled
 - SMTP relay information ← The "trace" fields
 - Sender's IP address, or
 - Initial SMTP server's IP address
 - Email server information
 - The Message ID ← Clearly labeled

Email Tracing

Step 1: Finding the Sender's Email Address

Email Tracing

- What can you do with the sender's email address?
 - Depends on what type of email address it is:
 - Hotmail, Yahoo!, etc. (free services)
 - AOL, Earthlink (pay for service)
 - Business/Organization email addresses
 - e.g., john.smith@search.org

Finding the Sender's Email Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

~~Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)~~

From: John Doe <johndoe@yahoo.com>

← Line 7

Subject: Fwd: Free kittens?

To: jos@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]

Begins with "From"

Email Tracing

Step 2: Find the Originating IP Address

Find the Originating IP Address

- **Goals**
 - Determine to whom the IP address is assigned.
 - Usually a business of some sort.
 - Often an Internet Service Provider (ISP) in the case of email.
 - Can be an Internet café, library, or other public Internet access point.
 - Contact the entity it is assigned to.
 - From their records, determine the identity of the person using that address at the date and time in question.

Types of Logs

- Email server logs may record:
 - The IP address of the system used by the email sender.
 - The sender's user ID.
 - The host name of the sender's computer.
 - The Message-ID assigned to the email sent by the sender.

Types of Logs

- Connection logs
 - Each email server, generally, has the ability to track connections from other servers.
 - This is in addition to the SMTP logs.
 - These connections are often recorded in a server connection log.
 - These logs are volatile and may be overwritten if not protected.
 - These logs are kept at the server level and are not included in email headers.

Find the Originating IP Address

- Potential problems:
 - IP addresses can be statically or dynamically assigned
 - Static: assigned to you and you only
 - e.g., Some businesses, cable and DSL subscribers
 - Dynamic: you borrow it and it is shared among several users
 - e.g., dial-up subscribers and organizations running DHCP
 - Requires user activity logs, which can quickly disappear

Preservation (“Freeze”) Order

- 18 USC Sec. 2703(f)
 - <http://uscode.house.gov/usc.htm>

(f) Requirement to Preserve Evidence. –

(1) In general. - A provider of wire or electronic communication services or a remote computing service, upon the request of a governmental entity, shall take all necessary steps to preserve records and other evidence in its possession pending the issuance of a court order or other process.

(2) Period of retention. - Records referred to in paragraph (1) shall be retained for a period of 90 days, which shall be extended for an additional 90-day period upon a renewed request by the governmental entity.

Preservation (“Freeze”) Order

- Caveats:
 - ISP may attempt to notify the target about your actions.
 - The ISP may terminate the account.

Find the Originating IP Address

- Other potential problems:
 - Forged fields
 - All fields in an email header can be forged by the sender except for the fields added by servers not under the sender's control. Forgeable fields include:
 - From:
 - Bogus, additional "Received" fields not created by actual servers.
 - X-Originating-IP:
 - Message-ID:
 - Reply-To:

Find the Originating IP Address

- First, locate the SMTP relaying information called the “trace” fields.
 - This is a listing of all the servers that processed the message via SMTP from the first to the last.
 - Indicated by the “Received:” header fields.
 - The receiving SMTP server creates the field in the header.

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)

From: John Doe <johndoe@yahoo.com>

Subject: Fwd: Free kittens?

To: jqs@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]

Beginning line #9

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

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Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- Think of the trace fields as chain-of-custody records.
- The first “Received:” line is a record of the email being relayed for the first time via SMTP.

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- The second “Received:” line is a record of the email being relayed for the second time via SMTP.

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- The third “Received:” line is a record of the email being relayed for the third time via SMTP.

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); **Wed, 16 Oct 2002 09:03:22 -0700**

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; **Wed, 16 Oct 2002 09:08:38 -0800**

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; **Wed, 16 Oct 2002 09:03:21 PDT**

- Each line identifies the sending and receiving hosts with the words “from” and “by,” respectively.
 - The receiving hosts write these lines into the header.
- The receiving hosts also **timestamp** the exchange.

Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- **The first Received line is read as follows:**
This message was received from a computer identified by the IP address 207.93.64.85 by a computer identified as web12608.mail.yahoo.com on Wednesday, October 16, 2002, at 09:03:21 PDT, according to the clock of the receiving computer.

Find the Originating IP Address

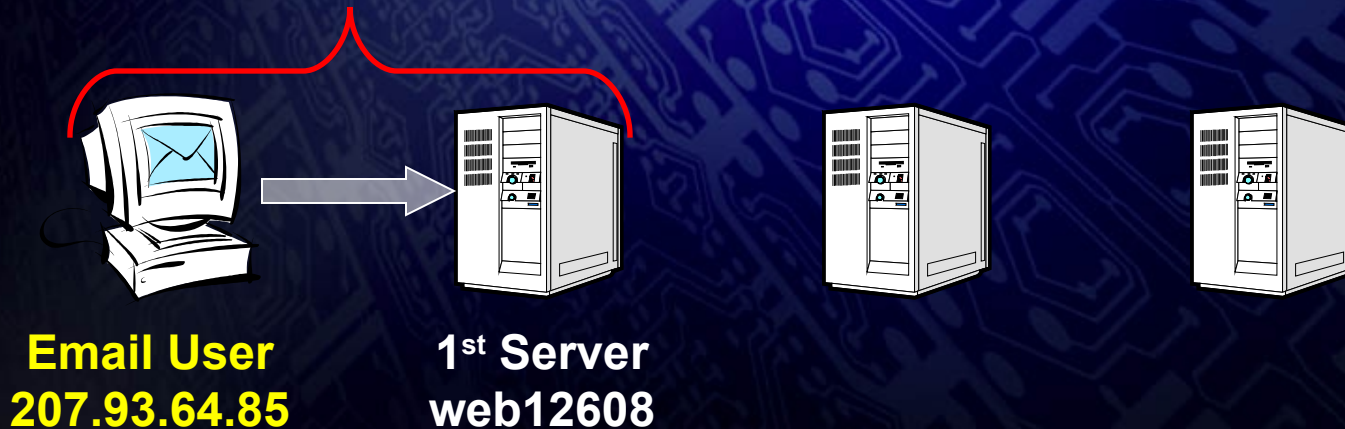
Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT



Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- **The second Received line is read as follows:**
This message was received from a computer identified as web12608.mail.yahoo.com with the IP address 216.136.173.231 by a computer identified as SEARCH.ORG on Wednesday, October 16, 2002, at 09:08:38 -0800, according to the clock of the receiving computer.

Find the Originating IP Address

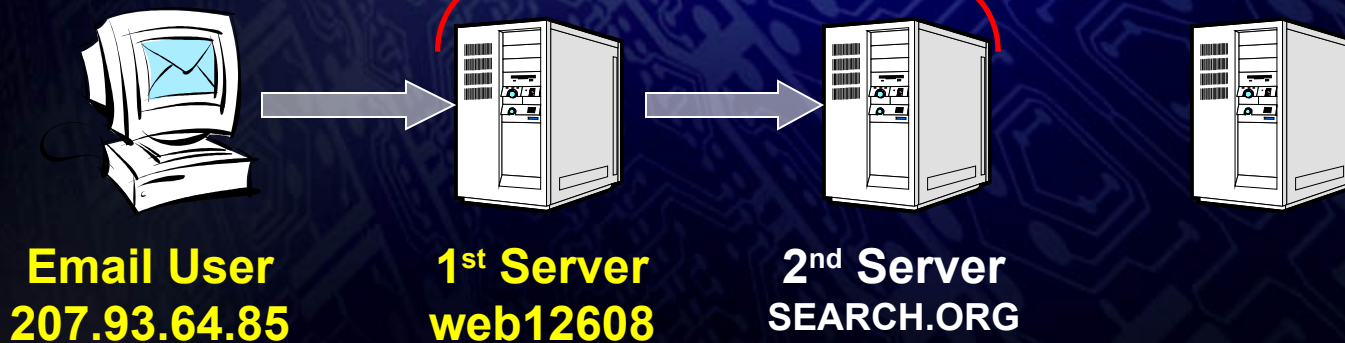
Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT



Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- **The third Received line is read as follows:**
This message was received from a computer identified as SEARCH.ORG with the IP address 64.162.18.2 by a computer identified as sgisrv1.search.org on Wednesday, October 16, 2002, at 09:03:22 –0700, according to the clock of the receiving computer.

Find the Originating IP Address

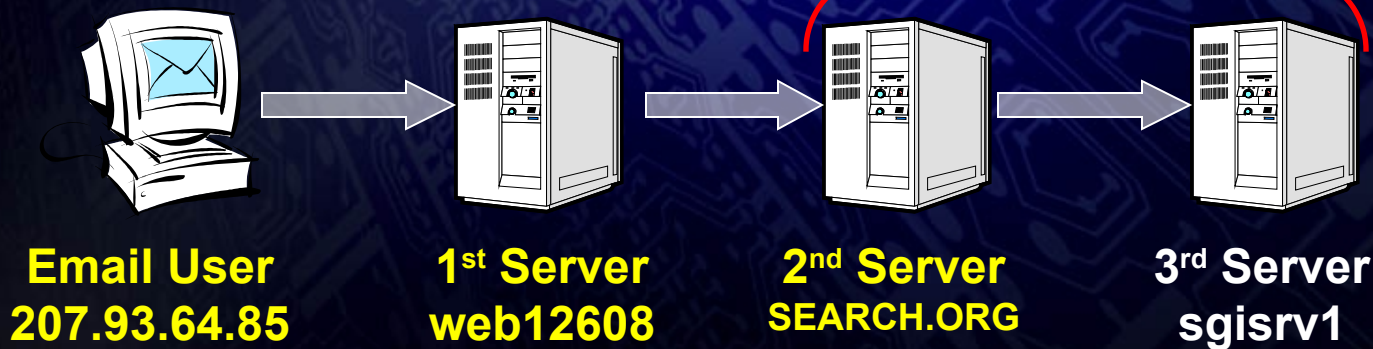
Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT



Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- The unbroken chain-of-custody is:
 - From 207.93.64.85 to web12608.mail.yahoo.com
 - From web12608.mail.yahoo.com to SEARCH.ORG
 - From SEARCH.ORG to sgisrv1.search.org
- 
- ```
graph LR; A[207.93.64.85] --> B[web12608.mail.yahoo.com]; B --> C[SEARCH.ORG]; C --> D[sgisrv1.search.org]
```

# Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- To identify the origins of an email, find the first (the bottom) Received record.
  - This is the email's entry into an SMTP environment.
  - It is the closest to identifying the sender's host as one can get from information available in a header.



# Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- Remember!
  - Bogus “Received” fields can be added.
  - Bogus fields are likely to be lower on the list.
  - Always check for a broken chain of custody.
  - There are real-world implementations of SMTP where the chain of custody is broken.

# Find the Originating IP Address

- Example of legitimate, broken chain of trace fields with Hotmail service.

Received: from search.org ([64.162.18.2]) by sgiserver1.search.org with SMTP  
(Microsoft Exchange Internet Mail Service Version 5.5.2650.21)

id PJMT6VX2; Thu, 26 Jul 2001 18:44:12 -0700

Received: from hotmail.com ([64.4.15.196]) by SEARCH.ORG

with ESMTP (IPAD 2.52) id 1832600; Thu, 26 Jul 2001 17:42:38 -0800

Received: from mail pickup service by hotmail.com with Microsoft SMTPSVC;  
Thu, 26 Jul 2001 18:43:43 -0700

Received: from 64.168.6.152 by lw10fd.law10.hotmail.msn.com with HTTP;  
Fri, 27 Jul 2001 01:43:42 GMT

This line should show “from”  
lw10fd.law10.hotmail.msn.com



# Find the Originating IP Address

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

- In this example, the computer identified by the IP address 207.93.64.85 is the point of origin into an SMTP environment.
  - It could be an IP address belonging to:
    - The ISP of the sender.
    - Their place of employment.
    - An Internet café, etc.

# Find the Originating IP Address

- To determine which organization is responsible for the administration of the originating IP address, use the WHOIS service of one of the Regional Internet Registries.





# Find the Originating IP Address

- In this example, the IP address 207.93.64.85 belongs to Earthlink (a common ISP).

Search results for: 207.93.64.85

```
OrgName: Earthlink Network Inc.
OrgID: ERDD
Address: 1375 Peachtree Street, Level A
City: Atlanta
StateProv: GA
PostalCode: 30309
Country: US

NetRange: 207.93.64.0 - 207.93.95.255
```

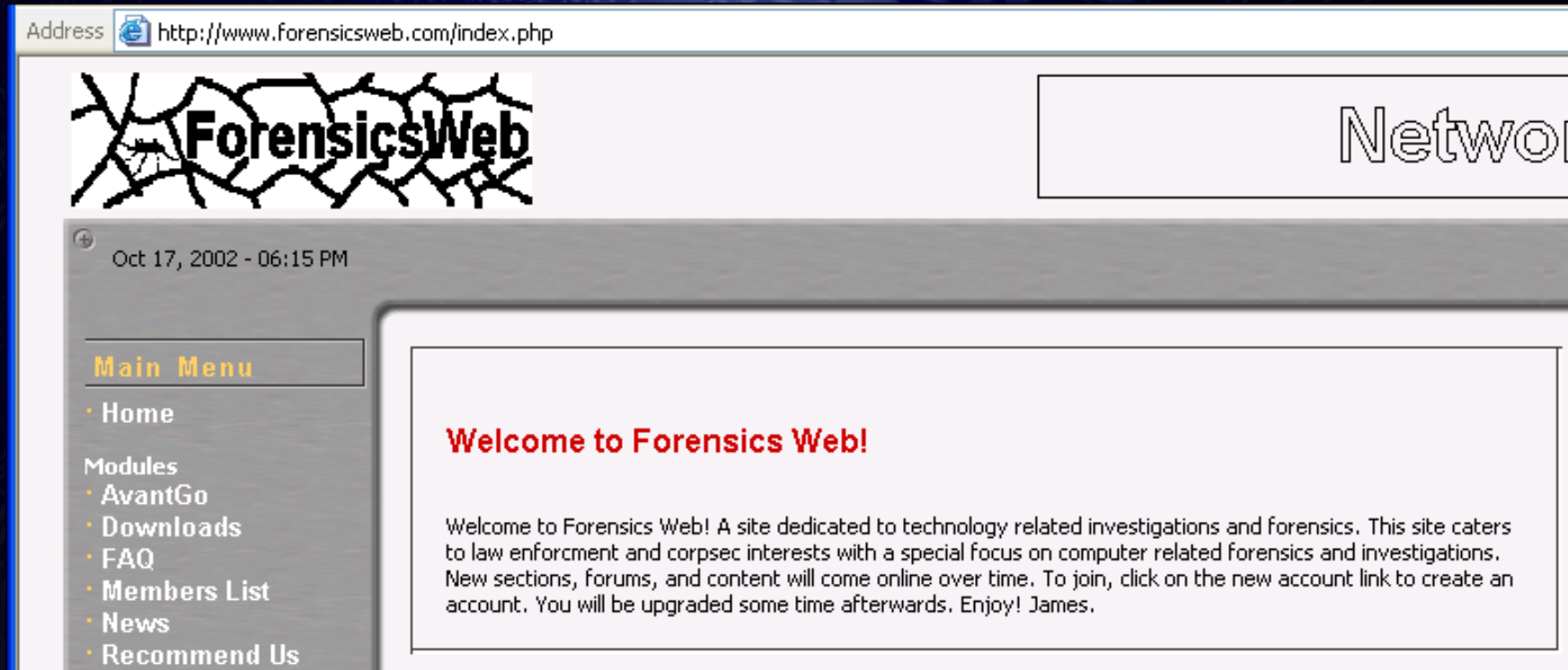
# Find the Originating IP Address

- Once the ISP is determined, legal contact information for law enforcement is available at <http://www.forensicsweb.com>
- Forensics Web only contains the contact information for companies that law enforcement is likely to deal with on a reoccurring basis.
  - If not on this list, can use information found in the RIR database.



# Find the Originating IP Address

- <http://www.forensicsweb.com>



# Find the Originating IP Address

- <http://www.forensicsweb.com>
- Legal contact for Earthlink

## EARTHLINK / MINDSPRING (08/03)

|                        |                                                        |
|------------------------|--------------------------------------------------------|
| Contact Name           | Subpoena Compliance Division, Esteban Martorell        |
| Online Service         | Earthlink Network Inc.                                 |
| Online Service Address | 1375 Peachtree Street, NE , Level A, Atlanta, GA 30309 |
| Voice Number           | (404) 748-6191                                         |
| Fax Number             |                                                        |

Subpoenas and search warrants are only accepted via US Certified Mail, UPS, FEDEX, and Airbourne Express



# Find the Originating IP Address

- Sometimes there will be a special trace field called X-Originating-IP that identifies the IP address assigned to the sender by their ISP for that session.

# Find the Originating IP Address

- Example of the X-Originating-IP field in a complete email header from Hotmail.

Received: from search.org ([64.162.18.2]) by sgiserver1.search.org with SMTP (Microsoft Exchange Internet Mail Service Version 5.5.2650.21) id PJMT6VX2; Thu, 26 Jul 2001 18:44:12 -0700  
Received: from hotmail.com ([64.4.15.196]) by SEARCH.ORG  
with ESMTP (IPAD 2.52) id 1832600; Thu, 26 Jul 2001 17:42:38 -0800  
Received: from mail pickup service by hotmail.com with Microsoft SMTPSVC;  
Thu, 26 Jul 2001 18:43:43 -0700  
Received: from 64.168.6.130 by lw10fd.law10.hotmail.msn.com with HTTP; Fri, 27 Jul 2001 01:43:42 GMT  
X-Originating-IP: [64.168.6.130]  
From: "Tea Party" <boston17@hotmail.com>  
To: jqs@search.org  
Bcc:  
Date: Fri, 27 Jul 2001 01:43:42 +0000  
Mime-Version: 1.0  
Content-Type: text/plain; format=flowed  
Message-ID: <F196q2lGhw3lqny6PO5000082f3@hotmail.com>  
X-OriginalArrivalTime: 27 Jul 2001 01:43:43.0012 (UTC) FILETIME=[91544640:01C1163D]



# Find the Originating IP Address

- The IP address in the X-Originating-IP field and the first Received field should be the same.

Received: from search.org ([64.162.18.2]) by sgiserver1.search.org with SMTP (Microsoft Exchange Internet Mail Service Version 5.5.2650.21) id PJMT6VX2; Thu, 26 Jul 2001 18:44:12 -0700

Received: from hotmail.com ([64.4.15.196]) by SEARCH.ORG  
with ESMTP (IPAD 2.52) id 1832600; Thu, 26 Jul 2001 17:42:38 -0800

Received: from mail pickup service by hotmail.com with Microsoft SMTPSVC;  
~~Thu, 26 Jul 2001 18:43:43 -0700~~

Received: from 64.168.6.130 by lw10fd.law10.hotmail.msn.com with HTTP; Fri, 27 Jul 2001 01:43:42 GMT

X-Originating-IP: [64.168.6.130]

~~From: "Tea Party" <boston17@hotmail.com>~~

To: jqs@search.org

Bcc:

Date: Fri, 27 Jul 2001 01:43:42 +0000

Mime-Version: 1.0

Content-Type: text/plain; format=flowed

Message-ID: <F196q2IGhw3lqny6PO5000082f3@hotmail.com>

X-OriginalArrivalTime: 27 Jul 2001 01:43:43.0012 (UTC) FILETIME=[91544640:01C1163D]

# Email Tracing

## Step 3: Find the Message-ID



# The Message-ID

- What is a Message-ID?
  - Every email is assigned a unique ID by the originating SMTP email server (the one listed next to “From” on the first Received line).

# The Message-ID

- Goals
  - Help detect spoofing attempts.
    - Example: the sender's email address.
  - Identify the originating SMTP server to obtains logs from.
  - Provide the ISP or email service provider with additional information to help locate the records needed to identify the sender.
    - It is easier, and usually faster, to search email server logs for the Message-ID.



# The Message-Id

- Message-IDs look very similar to email addresses, for example:

20021016160321.27540.qmail@web12608.mail.yahoo.com

- The information to the left of the “@” symbol is the unique identifier.
- Information to the right of the “@” identifies the domain to which the email server assigning the Message-ID belongs.
  - In this example, the email server belongs to the yahoo.com domain.

# The Message-ID

- Matching the domain information in the Message-ID to the domain information in the sender's email address will help determine the legitimacy of the sender's email address.
  - Not foolproof.
  - Matching information does not mean the sender's email address has not been forged.
  - If they do not match, it can be an indication that it has been forged.
  - Be familiar with the email service used.



# The Message-ID

- To find the Message-ID, look for a field in the header labeled Message-ID.

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

~~Received: from [207.03.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT~~

Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)

From: John Doe <johndoe@yahoo.com>

Subject: Fwd: Free kittens?

To: jqs@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]



# The Message-ID

- Domain information in the sender's address and Message-ID match.

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)

From: John Doe <johndoe@yahoo.com>

Subject: Fwd: Free kittens?

To: jqs@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]



# The Message-ID

- Additional information about some Message-IDs.

20021016160321.27540.qmail@web12608.mail.yahoo.com

- The numbers in this particular ID correspond to the date and time the ID was assigned.
  - Not all email servers do this.
- web12608 is likely the name of the email server. If so, this can be used to help locate the specific connection and message logs.

# The Message-ID

- Example SMTP log matching Message-ID to sender's IP address.

64.168.6.152, jqsmith, MAIL, 64.168.6.141, 000a01c273ef\$23ff7f40\$8206a840@mail.search.org

Message-ID

Email Server's IP Address

Host name of email server

Host name of sender's computer

Sender's IP Address



# Additional Header Information

- Another relevant field - Date:
  - Might indicate sender's location by time zone.

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 16 Oct 2002 09:03:22 -0700

Received: from web12608.mail.yahoo.com ([216.136.173.231]) by SEARCH.ORG with SMTP (IOA-IPAD 2.54) id 3548400; Wed, 16 Oct 2002 09:08:38 -0800

Message-ID: <20021016160321.27540.qmail@web12608.mail.yahoo.com>

Received: from [207.93.64.85] by web12608.mail.yahoo.com via HTTP; Wed, 16 Oct 2002 09:03:21 PDT

Date: Wed, 16 Oct 2002 09:03:21 -0700 (PDT)

From: John Doe <johndoe@yahoo.com>

Subject: Fwd: Free kittens?

To: jqs@search.org

MIME-Version: 1.0

Content-Type: multipart/mixed; boundary="0-1242707647-1034784201=:26523"

Return-Path: johndoe@yahoo.com

X-OriginalArrivalTime: 16 Oct 2002 16:03:22.0330 (UTC) FILETIME=[8D3C1BA0:01C2752D]

# Additional Header Information

- Helps determine which day's logs are needed.
- Can be used to determine the sender's general location by time zone if the information comes from their system clock.
  - Must be interpreted cautiously.
- The information in the Date field can come from different sources:
  - The date and time from the sender's computer, or
  - The date and time from the initial email server.



# Additional Header Information

- Whether the information in the Date field comes from the sender's computer clock or the email server's clock depends on the email service used. Using Hotmail for example:
  - If sent from the Hotmail web site, the timestamp in the date field comes from a Hotmail email server.
  - If sent using Outlook Express through the Hotmail service, it comes from the system clock of the sender's computer.

# Additional Header Information

- Determining where the Date field information comes from will require experimentation with the email service in question.
- It may not be possible to determine the timestamp source.
  - E.g., Outsiders do not have access to the email system.
- Always use caution when drawing a conclusion about any timestamp information in a header.
  - System clocks can be inaccurate.



# Additional Header Information

- A method for Hotmail: check for email client information in the header.

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 23 Oct 2002 10:53:19 -0700

Received: from hotmail.com ([216.33.240.88]) by SEARCH.ORG with ESMTP (IOA-IPAD 2.54) id 4171800; Wed, 23 Oct 2002 10:59:54 -0800

Received: from mail pickup service by hotmail.com with Microsoft SMTPSVC; Wed, 23 Oct 2002 10:53:19 -0700

X-Originating-IP: [64.162.18.159]

From: "TC" boston17@hotmail.com

To: jqs@search.org

Subject: Kittens for sale

Date: Wed, 23 Oct 2002 10:48:16 -0700

MIME-Version: 1.0

Content-Type: multipart/alternative; boundary="-----=\_NextPart\_000\_000A\_01C27A81.B1743CB0"

X-Priority: 3

X-MSMail-Priority: Normal

X-Mailer: Microsoft Outlook Express 6.00.2600.0000

~~X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000~~

Message-ID: OE31FFsAitVGeSJIEoP00003f05@hotmail.com

X-OriginalArrivalTime: 23 Oct 2002 17:53:19.0069 (UTC) FILETIME=[1216D8D0:01C27ABD]

Return-Path: boston17@hotmail.com

Email client  
information



# Additional Header Information

- A method for Hotmail: check for email client information in the header.

Microsoft Mail Internet Headers Version 2.0

Received: from SEARCH.ORG ([64.162.18.2]) by sgisrv1.search.org with Microsoft SMTPSVC(5.0.2195.3779); Wed, 23 Oct 2002 10:54:16 -0700

Received: from hotmail.com ([216.33.241.172]) by SEARCH.ORG with ESMTP (IOA-IPAD 2.54) id 4172300; Wed, 23 Oct 2002 11:00:51 -0800

Received: from mail pickup service by hotmail.com with Microsoft SMTPSVC; Wed, 23 Oct 2002 10:54:15 -0700

Received: from 64.162.18.159 by lw8fd.law8.hotmail.msn.com with HTTP; Wed, 23 Oct 2002 17:54:15 GMT

X-Originating-IP: [64.162.18.159]

From: "T C" boston17@hotmail.com

To: jqs@search.org

Bcc:

Subject: Kittens for sale

Date: Wed, 23 Oct 2002 17:54:15 +0000

Mime-Version: 1.0

Content-Type: text/plain; format=flowed

Message-ID: F172w1rbwsyK9hZIVhi0000674f@hotmail.com

X-OriginalArrivalTime: 23 Oct 2002 17:54:15.0561 (UTC) FILETIME=[33C2D790:01C27ABD]

Return-Path: boston17@hotmail.com

No email client  
information, but  
HTTP specified.